

Change Ubuntu EC2 SSH Port from 22 to Custom Port 6421

Objective

Change the SSH port on an Ubuntu EC2 instance from the default port 22 to a custom port 6421.

Important: Keep port 22 allowed in the AWS Security Group until the new port 6421 has been successfully tested.

Step 1: Allow Custom Port 6421 in AWS Security Group

AWS Console → **EC2** → **Instance** → **Security** → **Security Groups** → **Inbound Rules**

Add:

Type: Custom TCP

Port: 6421

Source: Your IP / Required CIDR

Keep the existing port 22 rule.

Example:

TCP 22 → Allowed temporarily

TCP 6421 → Allowed

Step 2: Connect to the EC2 Instance

Connect using the existing SSH port 22.

```
ssh -i your-key.pem ubuntu@<SERVER-IP>
```

Keep this SSH session open throughout the change.

Step 3: Check the Current SSH Port

```
sudo ss -lntp | grep ssh
```

Expected initially:

```
LISTEN 0 4096 0.0.0.0:22 0.0.0.0:* users:("sshd",...)  
LISTEN 0 4096 [::]:22 [::]:* users:("sshd",...)
```

Step 4: Back Up SSH Configuration

```
sudo cp /etc/ssh/sshd_config /etc/ssh/sshd_config.bak
```

Step 5: Change SSH Port in sshd_config

Open the SSH server configuration:

```
sudo nano /etc/ssh/sshd_config
```

Find:

```
#Port 22
```

Change it to:

```
Port 6421
```

Save and exit:

```
Ctrl + O
```

```
Enter
```

```
Ctrl + X
```

Step 6: Verify SSH Configuration

Check the configured port:

```
sudo grep -nE '^[[:space:]]*Port[[:space:]]+' /etc/ssh/sshd_config
```

Expected:

```
36:Port 6421
```

Check the effective SSH configuration:

```
sudo sshd -T | grep '^port'
```

Expected:

```
port 6421
```

Validate the SSH configuration:

```
sudo sshd -t
```

If there is **no output**, the configuration syntax is valid.

Step 7: Check systemd SSH Socket

Ubuntu may use systemd socket activation. Check:

```
sudo systemctl status ssh.socket --no-pager
```

If it shows:

```
Listen: 0.0.0.0:22  
        [::]:22
```

the systemd socket must also be changed.

Check the configuration:

```
sudo systemctl cat ssh.socket
```

If the socket contains:

```
ListenStream=0.0.0.0:22
```

```
ListenStream=[::]:22
```

create a systemd override.

Step 8: Create systemd Override Directory

```
sudo mkdir -p /etc/systemd/system/ssh.socket.d
```

Step 9: Create the Override File

```
sudo nano /etc/systemd/system/ssh.socket.d/override.conf
```

Add:

```
[Socket]
```

```
ListenStream=
```

```
ListenStream=0.0.0.0:6421
```

```
ListenStream=[::]:6421
```

Save:

Ctrl + O

Enter

Ctrl + X

Verify:

```
sudo cat /etc/systemd/system/ssh.socket.d/override.conf
```

Expected:

[Socket]

ListenStream=

ListenStream=0.0.0.0:6421

ListenStream=[::]:6421

Step 10: Reload systemd

```
sudo systemctl daemon-reload
```

Check the effective socket configuration:

```
sudo systemctl cat ssh.socket
```

The override should be shown under:

```
/etc/systemd/system/ssh.socket.d/override.conf
```

Step 11: Restart SSH Socket

```
sudo systemctl restart ssh.socket
```

Check the socket:

```
sudo systemctl status ssh.socket --no-pager
```

Expected:

Listen: 0.0.0.0:6421 (Stream)

[::]:6421 (Stream)

Step 12: Verify SSH Listening Port

```
sudo ss -lntp | grep -E '.*:22:.*:6421'
```

Expected:

```
LISTEN ... 0.0.0.0:6421 ...  
LISTEN ... [::]:6421 ...
```

There should be **no port 22 listener**.

You can also check specifically:

```
sudo ss -lntp | grep 6421
```

Step 13: Check SSH Service

```
sudo systemctl status ssh --no-pager
```

Expected:

Active: active (running)

The service log should show:

```
Server listening on 0.0.0.0 port 6421.  
Server listening on :: port 6421.
```

Step 14: Test SSH Using the New Port

Do not close the existing SSH session.

Open a **new terminal** on your local machine.

Connect using port **6421**:

```
ssh -i your-key.pem -p 6421 ubuntu@<SERVER-IP>
```

Example:

```
ssh -i airflow-test.pem -p 6421 ubuntu@<SERVER-IP>
```

Confirm that you can successfully log in.

Step 15: Remove Port 22 from AWS Security Group

Only after successfully connecting through port 6421:

AWS Console → **EC2** → **Security Groups** → **Inbound Rules**

Remove:

SSH
TCP
22

Keep:

Custom TCP
TCP
6421
Your IP / Required CIDR

Step 16: Final Verification

Check on the server:

```
sudo ss -lntp | grep ':22'
```

Expected:

No output

Check port 6421:

```
sudo ss -lntp | grep ':6421'
```

Expected:

LISTEN ... 0.0.0.0:6421 ...

LISTEN ... [::]:6421 ...

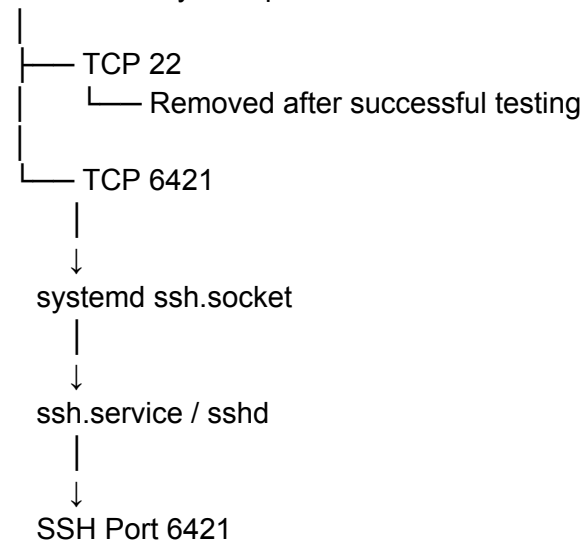
Test SSH:

ssh -i your-key.pem -p 6421 ubuntu@<SERVER-IP>

Port 22 should no longer be reachable through the Security Group.

Final Configuration

AWS Security Group



Important Safety Rule

Never close the original SSH session until a new SSH connection using port 6421 has been successfully tested.

If the new connection fails, the original session can be used to troubleshoot and restore SSH access.